BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

Section 'A' (Multiple Choice Questions) — MCQs 1–20 of 40 (scan incomplete)

Maximum Marks: 40

Note: Section A carries 40 MCQs in the original paper, but the source scan ends at Q20 — MCQs 21–40 are not available in the scan and are honestly omitted here. (Options in the original are separated by star marks, shown here as A–D in printed order.)

Note: Time: 3 Hours — Maximum Marks: 100

Q1. Choose the correct answer for each from the given options:

Q1. If $z = 4 + 3i$ then $z + \bar{z}$:

(A) 6 (B) 10

(C) 8 (D) 9i

Q2. $(1 + i)^2 =$: (where $i = \sqrt{-1}$):

(A) $-2i$ (B) $2i$

(C) 2 (D) -2

Q3. If $A = [-7]$, then $|A|$

(A) 7 (B) -7

(C) 1 (D) 0

Q4. If $|\bar{a} \times \bar{b}| = |\bar{a} \cdot \bar{b}|$ then the angle between $\bar{a}$ and $\bar{b}$ is:

(A) 0 (B) $\pi/2$

(C) $\pi/4$ (D) π

Q5. $\Sigma(n=3 \text{ to } 20)\ n^0$:

(A) 1 (B) 18

(C) 19 (D) 20

Q6. The factorial form of 12.11.10 is:

(A) 12!/9!
(B) 12!
(C) (12 choose 9)
(D) (12!).(9!)

Q7. No term of a G.P. is:

(A) 0
(B) 1
(C) Negative
(D) ∞

Q8. $\sin(\alpha/2) =$:

(A) $\sqrt{((s-b)(s-c)/bc)}$
(B) $\sqrt{((s-c)(s-a)/ca)}$
(C) $\sqrt{((s-a)(s-b)/ba)}$
(D) $\sqrt{(s(s-a)(s-b)(s-c))}$

Q9. Solution space of the linear inequality $2x + 3y \leq 6 \; \forall \; x,y \in R$ includes:

(A) All points above the line
(B) All points on and below the line
(C) All points below the line
(D) All points on and above the line

Q10. A circle, which passes through all vertices of a triangle is called the:

(A) tri-circle
(B) e-circle
(C) incircle
(D) circum circle

Q11. If K is a scalar and A, B, C are the square matrices of the same order, then $(kABC)^t =$:

(A) $kA^tB^tC^t$
(B) $kC^tB^tA^t$
(C) $k(BA)^t$
(D) $k^t(A.B)$

Q12. If $(a-b)/(b-c) = a/b$, then a, b and c are in:

(A) A.P
(B) G.P
(C) H.P
(D) A.G.P

Q13. If $a_n = 5 - 3n + 2n^2$, then $a_{2n} =$:

(A) $5 - 6n + 2n^2$
(B) $5 - 6n + 4n^2$
(C) $5 + 6n + 4n^2$
(D) $5 - 6n + 8n^2$

Q14. In tossing a coin, the probability of getting the tail is:

(A) 1/2
(B) 1/3
(C) 1/4
(D) 1/6

Q15. $1 + x + x^2 + x^3 + \ldots\ldots = :$

(A) $(1 + x)^{-1}$
(B) $(1 - x)^{-2}$
(C) $(1 + x)^{-2}$
(D) $(1 - x)^{-1}$

Q16. The graph of a linear function is:

(A) curve (B) parabola

(C) straight line (D) triangle

Q17. If f and g are equal, where $f(x) = 7x - 4$ and $g(x) = x$, then $x =$:

(A) 1/3 (B) 2/3

(C) 1 (D) 4/3

Q18. $\cos(2\pi + \theta) =$:

(A) $\sin\theta$ (B) $-\sin\theta$

(C) $\cos\theta$ (D) $-\cos\theta$

Q19. If $y = \sin x$, then its range is:

(A) $-1 \leq y \leq 1$ (B) R

(C) $0 \leq y \leq \pi$ (D) $-\pi/2 \leq y \leq \pi/2$

Q20. The solution of $\sqrt{2}\sin\theta - 1 = 0$ in the interval $[\pi/2, \pi]$ is:

(A) $\pi/4$ (B) $5\pi/4$

(C) $3\pi/4$ (D) $7\pi/4$

Section 'B' (Short-Answer Questions)

Maximum Marks: 40

> Note: Attempt any Ten part questions in all. All questions carry equal marks. (i.e. 4 marks of each part)

Q2.

(i) Solve the quadratic equation: $2z^2 + 3z + 5 = 0$ by completing the squares where z is a complex number.

(ii) Find the real numbers x, y, z such that matrix A is Hermiation matrix, where A = [[3, x+2i, yi], [3−2i, 0, 1+zi], [yi, i−xi, −1]]

(iii) Without expanding determinants, prove that $|\alpha\ \beta\gamma\ \alpha\beta\gamma;\ \beta\ \gamma\alpha\ \alpha\beta\gamma;\ \gamma\ \alpha\beta\ \alpha\beta\gamma| = |\alpha\ \alpha^2\ \alpha^3;\ \beta\ \beta^2\ \beta^3;\ \gamma\ \gamma^2\ \gamma^3|$

(iv) Find λ if the vectors $\hat{i} + \hat{j} + 2\hat{k}$, $\lambda\hat{i} - \hat{j} + \hat{k}$ and $3\hat{i} - 2\hat{j} - \hat{k}$ are coplanar.

(v) Find n so that (x^{n+5}+y^{n+5})/(x^{n+1}+y^{n+4}) may be A.M between x and y.

(vi) Find the sum of the series 3 + 6 + 15 + 42 + 123 + ...to n terms.

(vii) In how many ways can two English books, three chemistry books and four physics books be arranged on a shelf so that all the books of same subject are together?

(viii) Prove the proposition by mathematical induction for every positive integer n : $2 + 4 + 6 + \ldots\ldots + 2n = n(n + 1)$

(viii-OR) Write in the simplified form the term independent of x in the expansion of $(2x - 1/\sqrt{x})^9$.

(ix) Find the measure of the largest angle in ΔABC with a = 10cm, b = 20cm and c = 26cm.

(x) Solve the following linear programming problem by graphical method when $x \leq 0$, $y \geq 0$ maximize the objective function $Z(x,y) = 10x + 11y$, subject to constraints $2x + 3y \leq 8$; $6x + 3y \leq 10$.

(xi) Find the point of intersection of the function $f(x) = x + 4$ and $g(x) = x^2 - 6x + 10$ graphically.

(xi-OR) By using graph solve the equation $\cos\theta = 0$ for the interval $-\pi/2 \leq \theta \leq \pi/2$.

(xii) Express $15\sin\theta + 8\cos\theta$ in the form of $r \sin(\theta + \varphi)$ where θ and φ are in first quadrant.

(xiii) Prove that $r_1r_2 + r_2r_3 + r_3r_1 = s^2$.

(xiv) Show that: $\text{Tan}^{-1}(3/4) + \text{Tan}^{-1}(3/5) - \text{Tan}^{-1}(8/19) = \pi/4$.

Section 'C' (Detailed Answer Questions)

Maximum Marks: 40

Note: Attempt any FIVE questions from this section. All questions carry equal marks.

Q3.

Solve the following non-homogenous system of linear equations using Gauss Jordan − Method: $x - y + 4z = 4$; $2x + 2y - z = 2$; $3x - 2y + 3z = -3$

Q4.

A room has 3 lamps from a collection of 11 light bulbs of which 6 are no good, a person selects 3 at random and puts them in the sockets. What is the probability that he will light?

Q5.

If $1/x = 2/5 + (1-3)/2!\ (2/5)^2 + (1-3-5)/3!\ (2/5)^3 + \ldots\ldots$ then show that $4x^2 - 2x - 1 = 0$.

Q6.

If $f : R \to R$ is the function defined by $f(x) = 5x + 7$. Find $f^{-1}(x)$ and verify that $f^{-1}[f(x)] = x$.

Q7.

Prove that $\cos20° \cos40° \cos60° \cos80° = 1/16$ by using trigonometric formulae:

Q8.

Find the general solution of the trigonometric equation $\sin 4\theta - \sin 2\theta - \cos 3\theta = 0$ and verify the solution.

Q9.

A, B, C are the points $\bar{a}$, $\bar{b}$ and $2\bar{a} + \bar{b}$ respectively. D divides AC in 2 : 3 and E divides BD in 4 : 1. Find the position vector of E.